

# Understanding the Link Between Resilience and Physical Activity Among Adolescents



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## Introduction

- Adolescent physical activity (PA) is vital for long-term health, yet many youth fail to meet recommended guidelines
- Resilience may help youth maintain PA despite unsupportive environments, but research on its role in moderating neighborhood safety perceptions and PA participation is limited.
- Understanding this relationship can inform interventions that enhance environments, coping skills, and sustainable PA.

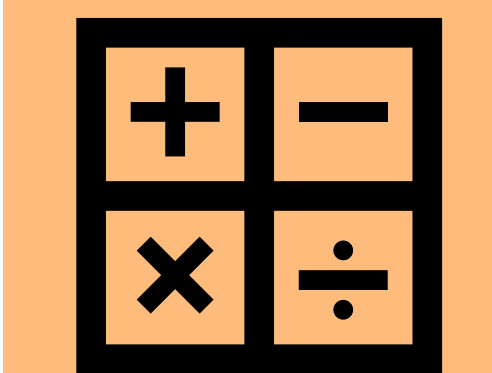
## Methods



Adolescents (n=41) aged 11-15 in Central Texas were recruited to complete a survey and wear Actigraph LEAPs for 7 days in Summer 2025.



Resilience was assessed using the Connor-Davidson Resilience Scale-10 in a baseline survey. Neighborhood safety perceptions were measured using the Neighborhood Environment Walkability Scale for Youth.



Generalized estimating equations and multilevel modeling were used to examine main effects of neighborhood safety on PA outcomes and test whether resilience moderated these relationships.

## Results

**Table 1&2:** Descriptive Statistics for Participants (n=41)

| Variables              | n (%) / M (SD) | Variables                                            | n (%) / M (SD) |
|------------------------|----------------|------------------------------------------------------|----------------|
| Age                    | 12.76 (.86)    | Valid accelerometer wear days per participant        | 6.68 (1.21)    |
| <b>Sex</b>             |                | Percent of days meeting physical activity guidelines | 16.5%          |
| Female                 | 10 (24.4%)     |                                                      |                |
| Male                   | 31 (75.6%)     | Traffic safety perception                            | 2.09 (0.46)    |
| <b>Race</b>            |                |                                                      |                |
| Asian                  | 4 (9.8%)       | Crime safety perception                              | 1.56 (0.71)    |
| Native American        | 1 (2.4%)       |                                                      |                |
| White                  | 34 (82.1%)     | Resilience (CD-RISC 10)                              | 34.93 (6.53)   |
| Multiracial            | 2 (4.8%)       |                                                      |                |
| <b>Ethnicity</b>       |                |                                                      |                |
| Hispanic or Latino/a/x | 15 (36.6%)     |                                                      |                |

**Table 3:** GEE Model: Predictors of Meeting Daily MVPA Guidelines (n=41).

| Predictor                   | β      | SE   | Wald  | OR   | p      |
|-----------------------------|--------|------|-------|------|--------|
| <b>Covariates</b>           |        |      |       |      |        |
| Intercept                   | -10.46 | 4.79 | 4.77  | —    | .03*   |
| Age                         | 0.55   | 0.35 | 2.48  | 1.74 | .12    |
| Sex                         | 1.08   | 0.45 | 5.68  | 2.95 | .02*   |
| <b>Main effects</b>         |        |      |       |      |        |
| Traffic safety              | 0.98   | 0.30 | 10.33 | 2.65 | .001** |
| Crime safety                | -1.50  | 0.53 | 8.05  | 0.22 | .005** |
| Resilience                  | 0.07   | 0.06 | 1.21  | 1.07 | .27    |
| <b>Interactions</b>         |        |      |       |      |        |
| Traffic safety × Resilience | 0.01   | 0.14 | 0.00  | 1.01 | .95    |
| Crime safety × Resilience   | 0.14   | 0.07 | 4.23  | 1.15 | .04*   |



Adolescents who reported **higher perceived traffic safety** were more likely to meet daily PA guidelines ( $\beta=0.98, p<.001$ ).



Adolescents who reported **lower perceived crime safety** were less likely to meet PA guidelines ( $\beta=-1.50, p<.01$ ).



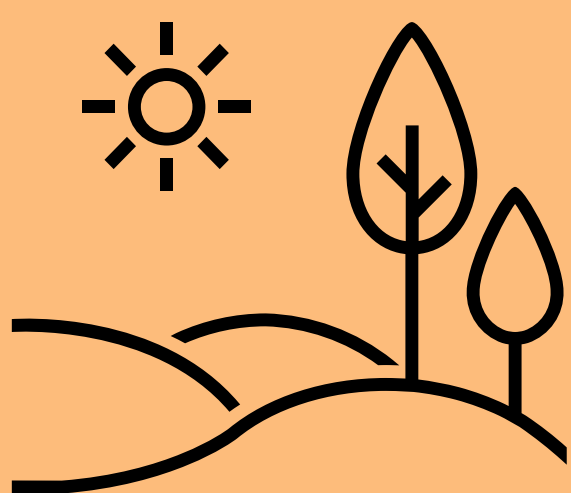
**Resilience moderated the relationship between perceived crime safety and PA**, suggesting resilient youth may be more likely to maintain PA in unsafe neighborhoods ( $\beta = 0.14, p = .04$ ).

This relationship was not shown with traffic safety and PA ( $p = .95$ ).

## Conclusions

Resilience may act as a protective factor that supports PA among adolescents despite environmental challenges.

Youth with stronger adaptive skills may be better able to be physically active even when barriers arise, particularly when they perceive their neighborhoods as unsafe.



These findings highlight the value of incorporating resilience-building strategies in PA promotion, as well as improving neighborhood safety.

Addressing both environmental conditions and individual coping capacity may better support PA among youth.

